ZIAD EMAD ALLAM

AI Automation & Full-Stack Engineer

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SUMMARY

Software engineer with 3+ years of experience in full-stack development, AI products, and test automation. Founder of ZEDX AI (20,000+ users, 490 premium subscribers, 2 B2B customers). At Elaraby Group, automated embedded-firmware testing in Python, replacing proprietary tooling and saving $70,000 per year. Stack: React/Next.js, Node.js, Python, LLM APIs.

EXPERIENCE

R&D Software Engineer | Elaraby Group 06/2024 - Present

- Built Python hardware-in-the-loop (HIL) test harnesses and DAQ pipelines for embedded firmware across washing machine, water heater, and microwave lines, with real-time signal analytics (NumPy, SciPy, FFT, filtering, anomaly detection).
- Replaced legacy proprietary and LabVIEW tooling with in-house Python automation, saving $70,000 per year and raising firmware test accuracy from 42% to 91%.
- Integrated Dassault 3DEXPERIENCE (PLM) for automated defect reporting and faster triage.

Full-Stack Developer (Freelance) | Upwork 11/2022 - 11/2025

- Delivered 10+ full-stack MERN projects for international clients with a 4.8/5 satisfaction rating.
- Built analytics dashboards that lifted user engagement 15% and CI/CD pipelines that cut manual deployment time 70%.

PROJECTS

ZEDX AI | Founder & Solo Developer | AI Interview Simulator zedx-ai.tech | GitHub

Next.js, React, TypeScript, Electron, Supabase, Groq (Whisper V3, Qwen, gpt-oss), ElevenLabs, Edge TTS

- Founded and solo-built a web and Electron desktop product for AI interview practice in 30 languages, growing to **20,000+ users, 490 premium subscribers, and 2 B2B customers**.
- Engineered a voice-to-voice pipeline: client-side VAD, Groq Whisper V3 STT (sub-second latency), Qwen/gpt-oss LLM inference over SSE, and a hybrid TTS router (ElevenLabs for Arabic, Edge TTS otherwise) that cuts TTS cost.
- Built load-balanced API key pools (14 for STT/LLM, 10 for TTS) with per-request shuffling and failover on errors.
- Injected each user's CV and job description into prompts for personalized interviews; strict-JSON LLM evaluation generates scorecards and PDF reports.
- Secured user data with Supabase Row Level Security, strict TypeScript models, and a prompt-injection guard against malicious resume content.

Sharp Inverter BLDC Washing Machine | HIL Test Automation Suite GitHub

Python, PyQt, nidaqmx, NumPy, SciPy, openpyxl, asyncio

- Designed a real-time HIL framework from scratch, replacing proprietary software ($40,000/yr saved) while processing 8-channel 10 kHz DAQ telemetry with no UI lag.
- Implemented finite state machines and DSP algorithms to validate motor agitation, closed-loop RPM dynamics, and 12 industrial fault-tree protocols; added oscilloscope views and automatic Excel diagnostic reports.

Heater & Microwave HIL Automation and AI Diagnostics GitHub

Python, PyQt6, nidaqmx, Pandas, SciPy, ML, Docker, CI/CD

- Migrated legacy LabVIEW lines to a Python framework with FFT-based DSP and ML predictive models: $30,000/yr cost reduction, test coverage 40% to 86%.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL
Frontend: React, Next.js, Tailwind CSS, Framer Motion, Electron
Backend: Node.js, Express, FastAPI, Flask, REST, WebSockets, JWT/RBAC
AI & LLM: LLM APIs (Groq, OpenAI, Anthropic), prompt engineering, STT/TTS pipelines, AI-assisted development (Claude, Cursor)
Data & Cloud: PostgreSQL, MySQL, SQLite, Supabase, AWS, Docker, CI/CD
Test Automation & Embedded: HIL testing, DAQ (nidaqmx), DSP (NumPy, SciPy, FFT), PyQt/PySide, state machines

EDUCATION

B.Sc. Computer and Information Systems | HICMIT, Tanta 10/2023 - 06/2027
GPA 3.5 | Data Structures, Algorithms, Software Development, Business and Project Management

Data Analysis & Visualization | The American University in Cairo (AUC) 05/2025 - 01/2026
CDAV101 final grade: A (4.0 scale)

GCI World Data Science | University of Tokyo, Matsuo-Iwasawa Laboratory 04/2026 - 08/2026
Fundamental theories and implementations of data science; passed final assessment

CERTIFICATIONS & LANGUAGES

Certifications: Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | DataCamp Certified Python Data Associate | Microsoft completion certificates (Azure OpenAI and Semantic Kernel agents, Cosmos DB, Functions, Container Apps, API Management, Key Vault, MSAL, Microsoft Graph, TypeScript) | micro1 Certified Software Engineer
Languages: Arabic (Native) | English (C1)